Research : Crowd Surveillance and Density Monitoring System

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Introduction

- A crowd is a large group of people gathered in public places such as railway stations, malls, events, and festivals.
- Managing and monitoring crowds manually using CCTV is difficult, time-consuming, and prone to human error.
- Overcrowding can lead to serious safety issues such as stampedes, accidents, and panic situations.
- With the growth of smart cities, there is a need for intelligent and automated surveillance systems.
- Computer Vision and Deep Learning enable automatic detection, counting, and density estimation of people in crowded scenes.
- AI-based crowd surveillance helps authorities take timely decisions and improve public safety.

Literature Survey

Paper	Focus	Core Method	Key Metric	Venue
P1: Chhajed et al.	Comprehensive review of AI crowd density systems	MCNN, FPN, GA-BP, GAN	MAE 1.2 (CDEM-M)	IEEE ICEI 2026
P2: Surendar et al.	Metro safety + onboard health IoT	Random Forest + IoT sensors	R^2 = 0.94, MAE = 2.83	IEEE IDCIoT 2026
P3: Joshna et al.	Weapon + crowd + face multi-task system	CNN + Triplet Loss + REST API	Accuracy 98.9%	IEEE ICETEMS 2026
P4: Chang	Crowd-crush prevention (CPI metric)	P2PNet + YOLOv8 + DBSCAN	Detects collapse 5–9 s early	ICACT 2026
P5: Tong et al.	RGB-Thermal crowd counting (SOTA)	VGG-19 + PVT + adaptive fusion	GAME(0) = 10.98 (SOTA)	IEEE TASE 2026

Problem Statement

Large gatherings in places like railway stations, festivals, malls, and public events can become dangerous due to overcrowding. Manual monitoring through CCTV is inefficient and cannot accurately estimate crowd size in real time.

There are main problems :

- Traditional methods fail due to **occlusion, lighting variation, and scale differences** in dense crowds.
- **Existing models struggle to generalize across different environments and camera angles**, reducing their reliability in real-world deployments.
- **Traditional surveillance systems do not work well in low-light or night conditions**, where visible cameras alone cannot detect crowds reliably.
- **Manual monitoring of CCTV cameras is inefficient and error-prone**, making it difficult to accurately detect and manage dense crowds in real time.

Objectives

- To develop an automated crowd surveillance system using deep learning.
- To detect and count people accurately in crowded scenes from CCTV videos.
- To estimate crowd density in real time for better monitoring.
- To handle challenges like occlusion, scale variation, and lighting changes.
- To reduce manual effort in surveillance and improve monitoring efficiency.
- To support public safety by identifying overcrowded and risky areas.
- To make the system suitable for smart city and public place surveillance.

Proposed Methodology

Our system uses **Computer Vision + Deep Learning**.

- **Steps Involved**
  - Input video/image from surveillance camera
  - Preprocessing and resizing
  - Person detection using deep learning models
  - Counting and density estimation
  - Performance evaluation
- **Models Used**
  - **Mobile Net-SSD** – Fast & lightweight
  - **YOLOv8** – High accuracy & real-time detection
  - **Mask R-CNN** – Detailed segmentation

System Architecture & workflow

1. Dataset Input

- CCTV video streams / surveillance footage
- Public crowd datasets (ShanghaiTech, UCF-CC-50, Mall Dataset)
- Frames extracted from video for processing
- Data diversity includes varying lighting, density, angles, occlusion

2. Pre-Processing Layer

- Frame extraction from video stream
- Image resizing & normalization
- Noise reduction & contrast enhancement
- Data augmentation (flip, rotate, scale)
- Annotation (bounding boxes / density maps)

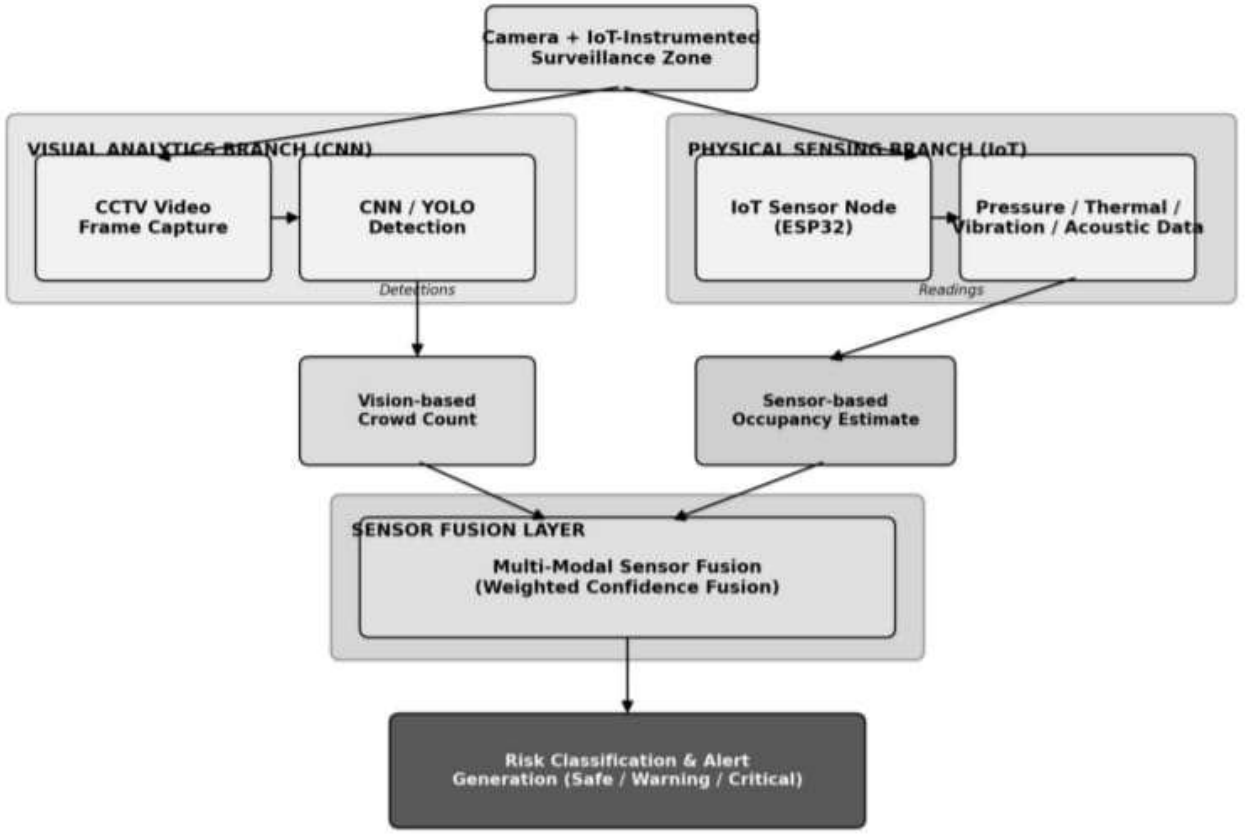
3. Detection & Analysis Models

- **MobileNet-SSD** → Fast detection for low-resource systems
- **YOLOv8** → Real-time accurate person detection
- **Mask R-CNN** → Precise segmentation in dense crowds

4. Output Layer

- Real-time crowd count display
- Density heatmap visualization
- Overcrowding alerts
- Risk level classification (Low / Medium / High)

Proposed System Architecture



: Proposed dual-channel system architecture (CNN visual branch + IoT sensing branch → sensor fusion → risk classification & alerts)

IoT Sensor Suite

Sensor	Quantity Sensed	Role in the System
IR Sensor	Present,movement or proximity of object.	Detects presence & approximate weight events at fixed floor points
Piezoelectric Footstep Sensor	Mechanical vibration from footstep impact	Counts discrete footstep events at a monitored doorway/strip
SW-420 Vibration Sensor	Ambient structural/floor vibration	Detects aggregate crowd movement & activity intensity in a zone

Implementation Details

- Live image is captured from CCTV cameras and converted into frames using OpenCV.
- Each frame is preprocessed (resizing, normalization) before being sent to the deep learning model.
- A trained model (YOLO / MobileNet-SSD / Mask R-CNN) is used to detect people in the scene.
- The number of detected people is counted to estimate crowd density in real time.
- The results are displayed on a dashboard with bounding boxes and crowd count.

Conclusion

- The proposed system successfully combines CNN/YOLO-based visual detection with a low-cost IoT sensor network for crowd monitoring in smart surveillance zones.

- The current IoT sensors (IR, piezoelectric, vibration) operate in a binary presence mode — they can only detect whether people are present (1) or not (0), not the actual headcount.

- This makes the sensor layer reliable for occupancy/activity confirmation and as a backup when the camera feed is blocked or lighting is poor, but it cannot independently estimate crowd density.

Future Work

- Add multi-zone thermal imaging (thermal camera arrays instead of single IR/PIR sensors) to estimate headcount from heat-signature clustering, especially useful at night.
- Calibrate acoustic and vibration sensors against known crowd sizes to derive an approximate people-count model instead of a simple activity/noise indicator.
- Improve sensor fusion algorithm to combine the upgraded quantitative sensor data with the CNN/YOLO vision count for higher accuracy and better confidence scoring.
- Real-time edge deployment — optimize the CNN/YOLO models and ESP32 sensor pipeline for faster on-device inference, reducing alert latency.

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Thank You